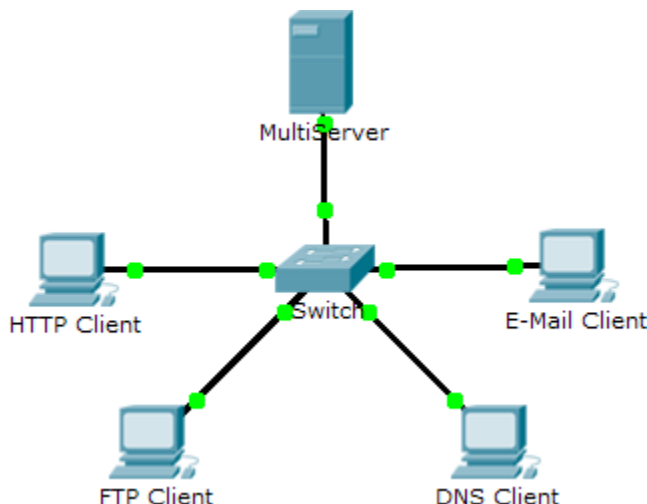


Packet Tracer Simulation - TCP and UDP Communications

Topology



Objectives

Part 1: Generate Network Traffic in Simulation Mode

Part 2: Examine the Functionality of the TCP and UDP Protocols

Background

This simulation activity is intended to provide a foundation for understanding the TCP and UDP in detail. Simulation mode provides the ability to view the functionality of the different protocols.

As data moves through the network, it is broken down into smaller pieces and identified in some fashion so that the pieces can be put back together. Each of these pieces is assigned a specific name (protocol data unit [PDU]) and associated with a specific layer. Packet Tracer Simulation mode enables the user to view each of the protocols and the associated PDU. The steps outlined below lead the user through the process of requesting services using various applications available on a client PC.

This activity provides an opportunity to explore the functionality of the TCP and UDP protocols, multiplexing and the function of port numbers in determining which local application requested the data or is sending the data.

Part 1: Generate Network Traffic in Simulation Mode

Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.

Perform the following tasks to reduce the amount of network traffic viewed in the simulation.

- Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.
- Enter the **ping 192.168.1.255** command. This will take a few seconds as every device on the network responds to **MultiServer**.
- Close the **MultiServer** window.

Step 2: Generate web (HTTP) traffic.

- Switch to Simulation mode.
- Click **HTTP Client** and click the **Desktop** tab > **Web Browser**.
- In the URL field, enter **192.168.1.254** and click **Go**. Envelopes (PDUs) will appear in the simulation window.
- Minimize, but do not close, the **HTTP Client** configuration window.

Step 3: Generate FTP traffic.

- Click **FTP Client** and click the **Desktop** tab > **Command Prompt**.
- Enter the **ftp 192.168.1.254** command. PDUs will appear in the simulation window.
- Minimize, but do not close, the **FTP Client** configuration window.

Step 4: Generate DNS traffic.

- Click **DNS Client** and click the **Desktop** tab > **Command Prompt**.
- Enter the **nslookup multiserver.pt.ptu** command. A PDU will appear in the simulation window.
- Minimize, but do not close, the **DNS Client** configuration window.

Step 5: Generate Email traffic.

- Click **E-Mail Client** and click the **Desktop** tab > **E Mail** tool.
- Click **Compose** and enter the following information:
 - To:** user@multiserver.pt.ptu
 - Subject:** Personalize the subject line
 - E-Mail Body:** Personalize the Email
- Click **Send**.
- Minimize, but do not close, the **E-Mail Client** configuration window.

Step 6: Verify that the traffic is generated and ready for simulation.

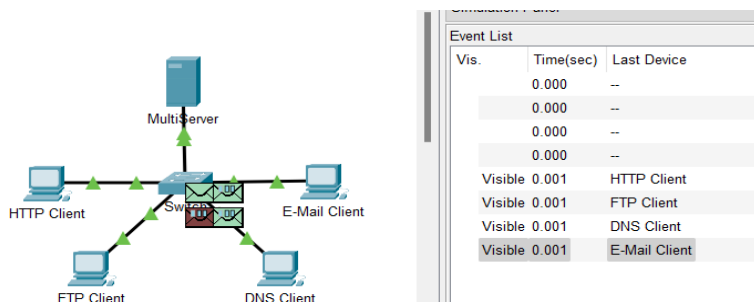
Every client computer should have PDUs listed in the Simulation Panel.

Part 2: Examine Functionality of the TCP and UDP Protocols

Step 1: Examine multiplexing as all of the traffic crosses the network.

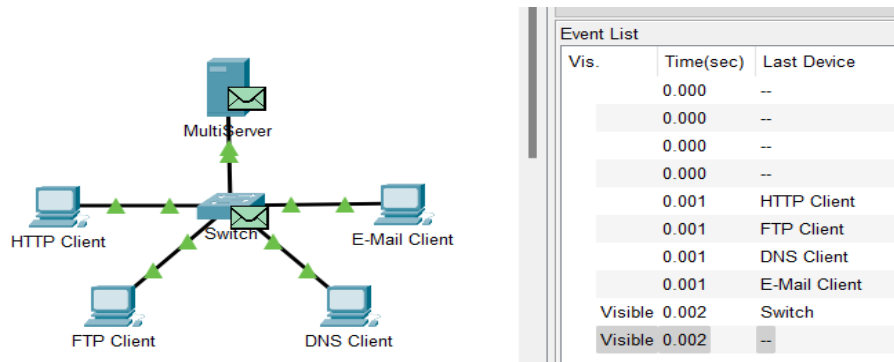
You will now use the **Capture/Forward** button and the **Back** button in the Simulation Panel.

- Click **Capture/Forward** once. All of the PDUs are transferred to the switch.



- b. Click **Capture/Forward** again. Some of the PDUs disappear. What do you think happened to them?

Ans: The PDUs that disappear have been stored in the switch.



- c. Click **Capture/Forward** six times. All clients should have received a reply. Note that only one PDU can cross a wire in each direction at any given time. What is this called?

Ans: Multiplexing

- d. A variety of PDUs appears in the event list in the upper right pane of the simulation window. Why are they so many different colors?

Ans: The different colors have represented the different protocols.

- e. Click **Back** eight times. This should reset the simulation.

Note: Do not click **Reset Simulation** any time during this activity; if you do, you will need to repeat the steps in Part 1.

Step 2: Examine HTTP traffic as the clients communicate with the server.

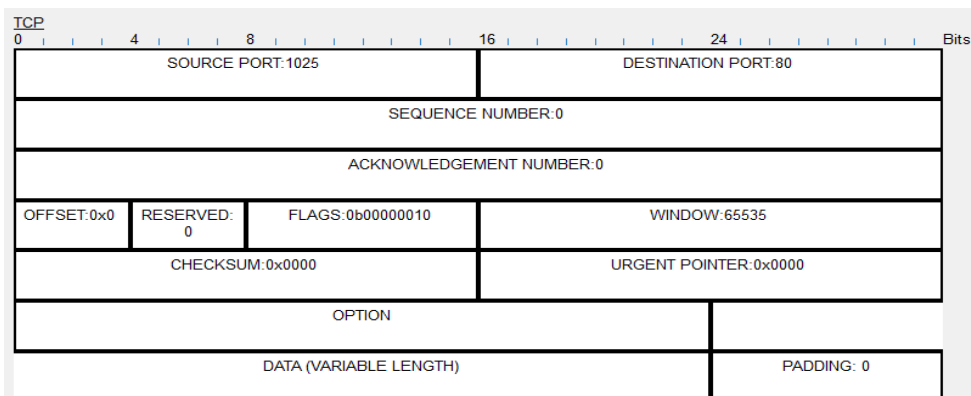
- a. Filter the traffic that is currently displayed to display only **HTTP** and **TCP** PDUs filter the traffic that is currently displayed:
- 1) Click **Edit Filters** and toggle the **Show All/None** check box.
 - 2) Select **HTTP** and **TCP**. Click anywhere outside of the Edit Filters box to hide it. The Visible Events should now display only **HTTP** and **TCP** PDUs.
- b. Click **Capture/Forward**. Hold your mouse above each PDU until you find one that originates from **HTTP Client**. Click the PDU envelope to open it.

- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

Ans: The section labeled is TCP.

Are these communications considered to be reliable?

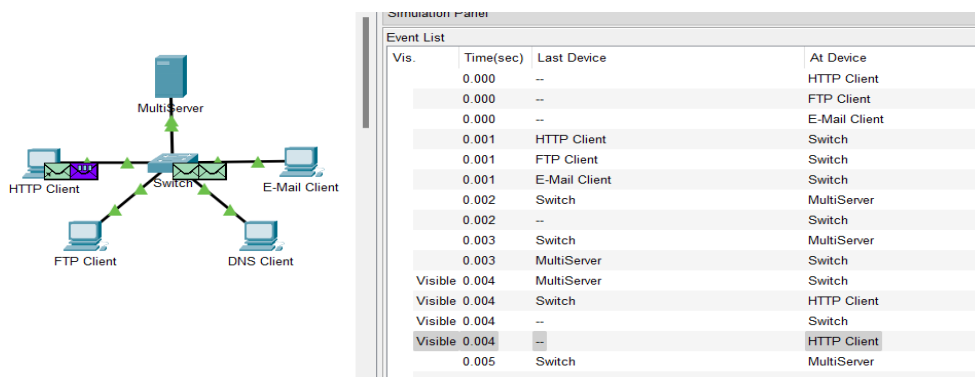
Ans: Yes.



- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

Ans: 1025, 80, 0, 0. SYN

- e. Close the PDU and click **Capture/Forward** until a PDU returns to the **HTTP Client** with a checkmark.



Packet Tracer Simulation - TCP and UDP Communications

- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

Ans: The port and sequence numbers are 80, 1025, 0, 1. SYN+ACK. The source and destination ports are reversed, and the acknowledgement number is 1. Besides, the SYN has changed to SYN+ACK.

TCP																Bits																			
0				4				8				16				24																			
SOURCE PORT:80												DESTINATION PORT:1025																							
SEQUENCE NUMBER:0																																			
ACKNOWLEDGEMENT NUMBER:1																																			
OFFSET:0x0				RESERVED: 0				FLAGS:0b00010010												WINDOW:16384															
CHECKSUM:0x0000																URGENT POINTER:0x0000																			
OPTION																																			
DATA (VARIABLE LENGTH)																								PADDING: 0											

- g. There is a second **PDU** of a different color, which **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the HTTP communication. Click this second PDU envelope and select **Outbound PDU Details**.
- h. What information is now listed in the TCP section? How are the port and sequence numbers different from the previous two PDUs?

Ans: The information listed in the TCP section is 1025, 80, 1, 1. PSH+ACK. Then, the source and destination ports are exchanged, and both sequence and acknowledgement numbers are 1.

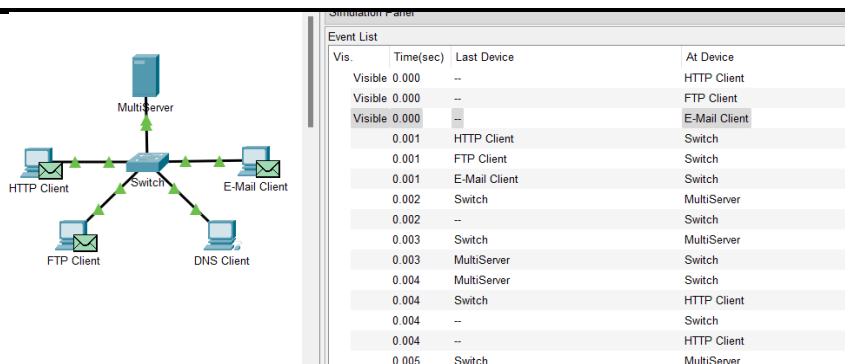
PDU Information at Device: HTTP Client																																																																																																			
OSI Model																																																																																																			
Outbound PDU Details																																																																																																			
PDU Formats																																																																																																			
DATA (VARIABLE LENGTH)																																																																																																			
TCP																																																																																																			
0 4 8 16 24 Bits																																																																																																			
SOURCE PORT:1025																																																		DESTINATION PORT:80																																																	
SEQUENCE NUMBER:1																																																																																																			
ACKNOWLEDGEMENT NUMBER:1																																																																																																			
OFFSET:0x0																									RESERVED:0										FLAGS:0b00011000																									WINDOW:65535																																							
CHECKSUM:0x0000																																																		URGENT POINTER:0x0000																																																	
OPTION																																																																																																			
DATA (VARIABLE LENGTH)																																																																											PADDING: 0																								

- i. Click **Back** until the simulation is reset.

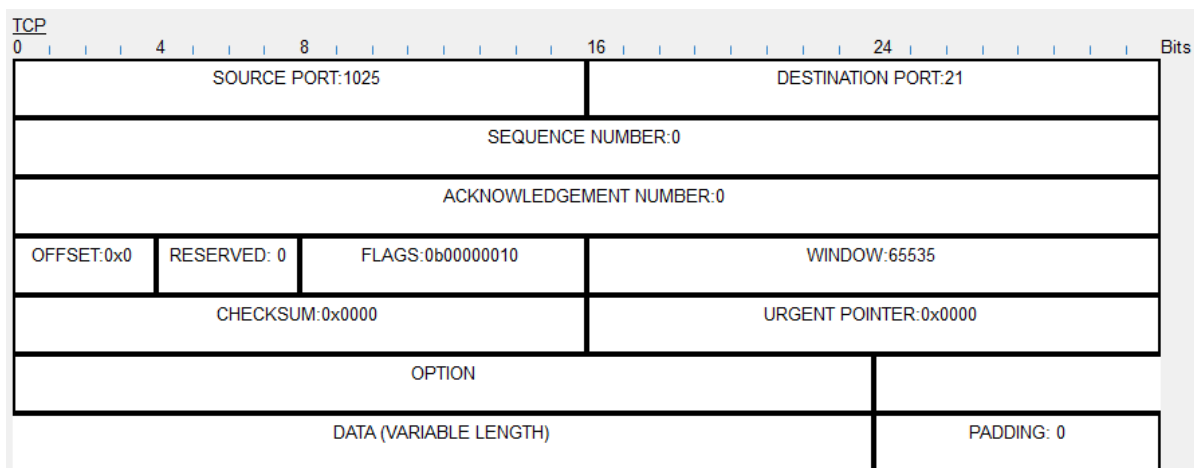
Step 3: Examine FTP traffic as the clients communicate with the server.

- a. In the Simulation Panel, change **Edit Filters** to display only **FTP** and **TCP**.
- b. Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **FTP Client**. Click that PDU envelope to open it.

Packet Tracer Simulation - TCP and UDP Communications



- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?
 Ans: TCP.



Are these communications considered to be reliable?

Ans: Yes.

- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

Ans: 1025, 21, 0, 0. SYN.

TCP																																Bits							
0				4				8				16				24																							
SOURCE PORT:1025																DESTINATION PORT:21																							
SEQUENCE NUMBER:0																																							
ACKNOWLEDGEMENT NUMBER:0																																							
OFFSET:0x0				RESERVED: 0				FLAGS:0b00000010								WINDOW:65535																							
CHECKSUM:0x0000																URGENT POINTER:0x0000																							
OPTION																																							
DATA (VARIABLE LENGTH)																								PADDING: 0															

- e. Close the PDU and click **Capture/Forward** until a PDU returns to the **FTP Client** with a checkmark.

- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

Ans: 21, 1025, 0, 1. SYN+ACK. The source and destination ports are exchanged, and the acknowledgement number is 1.

TCP																																Bits							
0				4				8				16				24																							
SOURCE PORT:21																DESTINATION PORT:1025																							
SEQUENCE NUMBER:0																																							
ACKNOWLEDGEMENT NUMBER:1																																							
OFFSET:0x0				RESERVED: 0				FLAGS:0b00010010								WINDOW:16384																							
CHECKSUM:0x0000																URGENT POINTER:0x0000																							
OPTION																																							
DATA (VARIABLE LENGTH)																								PADDING: 0															

- g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?

Ans: 1025, 21, 1, 1. ACK. The source and destination ports are exchanged, and both sequence and acknowledgement numbers are 1.

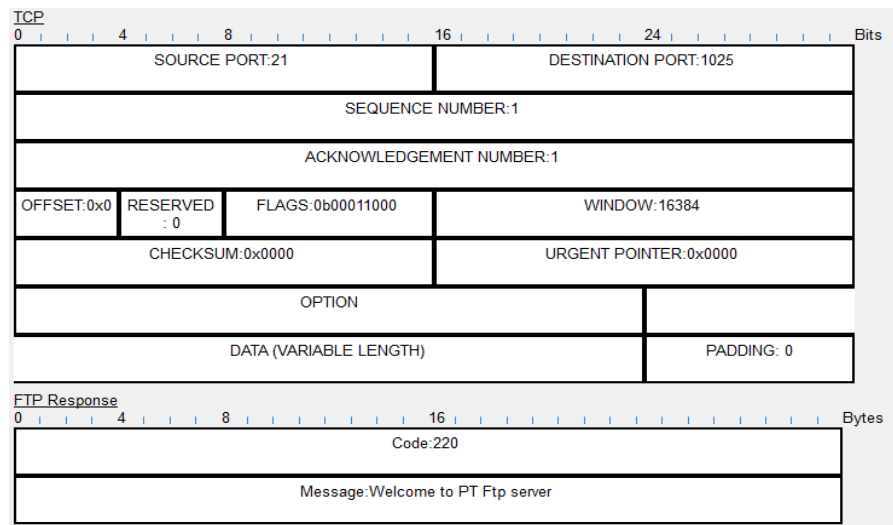
TCP																																Bits															
0								4								8								16								24															
SOURCE PORT:1025																DESTINATION PORT:21																															
SEQUENCE NUMBER:1																																															
ACKNOWLEDGEMENT NUMBER:1																																															
OFFSET:0x0				RESERVED: 0				FLAGS:0b00010000												WINDOW:65535																											
CHECKSUM:0x0000																URGENT POINTER:0x0000																															
OPTION																																															
DATA (VARIABLE LENGTH)																								PADDING: 0																							

- h. Close the PDU and click **Capture/Forward** until a second PDU returns to the **FTP Client**. The PDU is a

different color.

- i. Open the PDU and select **Inbound PDU Details**. Scroll down past the TCP section. What is the message from the server?

Ans: The message is "Welcome to PT Ftp server."

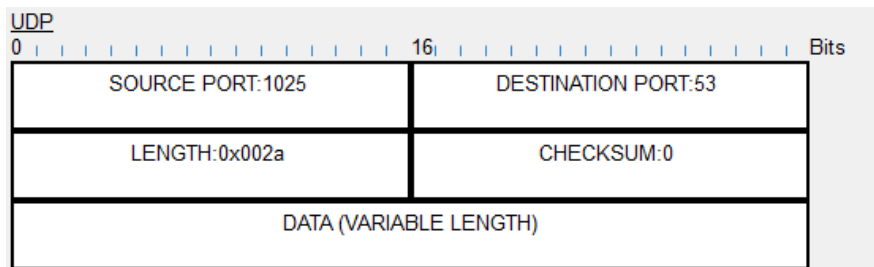


- j. Click **Back** until the simulation is reset.

Step 4: Examine DNS traffic as the clients communicate with the server.

- a. In the Simulation Panel, change **Edit Filters** to display only **DNS** and **UDP**.
- b. Click the PDU envelope to open it.
- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

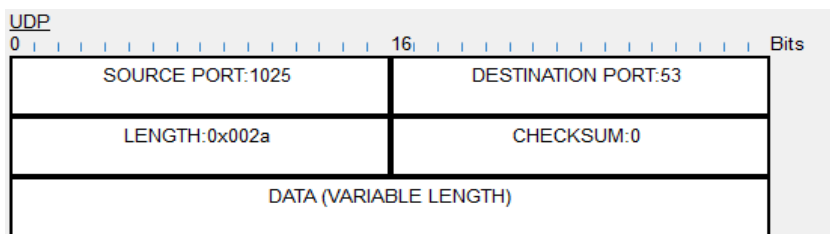
Ans: UDP.



Are these communications considered to be reliable? No

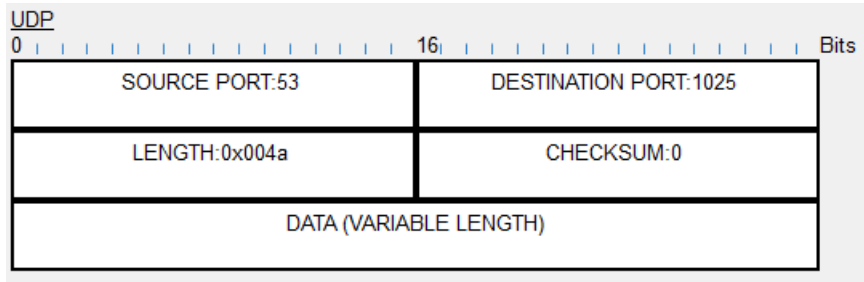
- d. Record the **SRC PORT** and **DEST PORT** values. Why is there no sequence and acknowledgement number?

Ans: 1025, 53. Because UDP does not need to establish a reliable connection.



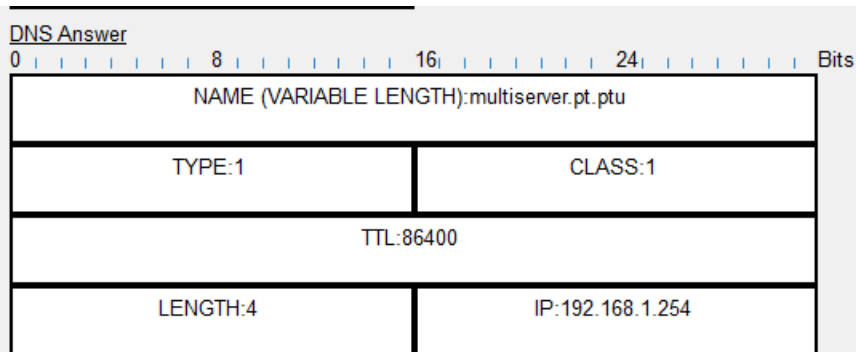
- e. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **DNS Client** with a checkmark.
- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

Ans: 53, 1025. The source and destination ports are reversed.



- g. What is the last section of the **PDU** called?

Ans: DNS ANSWER.



- h. Click **Back** until the simulation is reset.

Step 5: Examine email traffic as the clients communicate with the server.

- a. In the Simulation Panel, change **Edit Filters** to display only **POP3**, **SMTP** and **TCP**.

- b. Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **E-mail Client**. Click that PDU envelope to open it.
- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What transport layer protocol does email traffic use?

Ans: TCP.

Are these communications considered to be reliable?

Ans: Yes.

TCP																								Bits							
0		4		8		16		24																							
SOURCE PORT:1025												DESTINATION PORT:25																			
SEQUENCE NUMBER:0																															
ACKNOWLEDGEMENT NUMBER:0																															
OFFSET:0x0				RESERVED: 0				FLAGS:0b00000010				WINDOW:65535																			
CHECKSUM:0x0000												URGENT POINTER:0x0000																			
OPTION																															
DATA (VARIABLE LENGTH)																		PADDING: 0													

- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

Ans: 1025, 25, 0, 0. SYN.

- e. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **E-Mail Client** with a checkmark.
- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

Ans: 25, 1025, 0, 1. SYN+ACK. The source and destination ports are exchanged, and the acknowledgement number is 1.

TCP																Bits							
0		4		8						16				24									
SOURCE PORT:25										DESTINATION PORT:1025													
SEQUENCE NUMBER:0																							
ACKNOWLEDGEMENT NUMBER:1																							
OFFSET:0x0				RESERVE D: 0				FLAGS:0b00010010				WINDOW:16384											
CHECKSUM:0x0000										URGENT POINTER:0x0000													
OPTION																							
DATA (VARIABLE LENGTH)														PADDING: 0									

- g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?

Ans: 1025, 25, 1, 1. ACK. The source and destination ports are reversed, and both sequence and acknowledgement numbers are 1.

TCP		Bits	
0	4	8	16
SOURCE PORT:1025		DESTINATION PORT:25	
SEQUENCE NUMBER:1			
ACKNOWLEDGEMENT NUMBER:1			
OFFSET:0x0	RESERVE D: 0	FLAGS:0b00010000	WINDOW:65535
CHECKSUM:0x0000		URGENT POINTER:0x0000	
OPTION			
DATA (VARIABLE LENGTH)			PADDING: 0

- h. There is a second **PDU** of a different color that **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the email communication. Click this second PDU envelope and select **Outbound PDU Details**.

- i. How are the port and sequence numbers different from the previous two **PDUs**?

Ans: 1025, 25, 1, 1. PSH+ACK. The source and destination ports are reversed, and both sequence and acknowledgement numbers are 1.

TCP		Bits	
0	4	8	16
SOURCE PORT:1025		DESTINATION PORT:25	
SEQUENCE NUMBER:1			
ACKNOWLEDGEMENT NUMBER:1			
OFFSET :0x0	RESERVED VED: 0	FLAGS:0b0001100 0	WINDOW:65535
CHECKSUM:0x0000		URGENT POINTER:0x0000	
OPTION			
DATA (VARIABLE LENGTH)			PADDING: 0

- j. What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?
Ans: SMTP, POP3.

- k. Click **Back** until the simulation is reset.

Step 6: Examine the use of port numbers from the server.

- a. To see TCP active sessions, perform the following steps in quick succession:

- 1) Switch back to **Realtime** mode.

2) Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.

- b. Enter the **netstat** command. What protocols are listed in the left column?

What port numbers are being used by the server?

Ans: 21

```
C:\>netstat

Active Connections

    Proto Local Address          Foreign Address         State
    TCP    192.168.1.254:21      192.168.1.2:1025      ESTABLISHED
```

- c. What states are the sessions in?

Ans: ESTABLISHED.

- d. Repeat the **netstat** command several times until you see only one session still ESTABLISHED. For which service is this connection still open? Ans: FTP.

```
C:\>netstat

Active Connections

    Proto Local Address          Foreign Address         State
    TCP    192.168.1.254:21      192.168.1.2:1025      ESTABLISHED
C:\>netstat

Active Connections

    Proto Local Address          Foreign Address         State
    TCP    192.168.1.254:21      192.168.1.2:1025      ESTABLISHED
C:\>netstat

Active Connections

    Proto Local Address          Foreign Address         State
    TCP    192.168.1.254:21      192.168.1.2:1025      ESTABLISHED
C:\>netstat

Active Connections

    Proto Local Address          Foreign Address         State
    TCP    192.168.1.254:21      192.168.1.2:1025      ESTABLISHED
C:\>
```

Why doesn't this session close like the other three? (Hint: Check the minimized clients)

Ans: The server is waiting for a password from the client.

Suggested Scoring Rubric

Activity Section	Question Location	Possible Points	Earned Points
Part 2: Examine Functionality of the TCP and UDP Protocols	Step 1	15	
	Step 2	15	
	Step 3	15	
	Step 4	15	
	Step 5	15	
	Step 6	25	
Total Score		100	